Course Title: Cryptography and Network Security Lab	**Course Code**: BCSE309P
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EX1A – Classical Cryptographic Techniques

Code (Caesar Cipher):

Sender:

```
import java.util.*;

import java.io.*;

import java.net.*;

public class Caesar_Sender {

    private Socket socket = null;

    private DataInputStream input = null;

    private DataOutputStream output = null;

    private Scanner sc = null;

    public Caesar_Sender(String address, int port) {

        try {

            socket = new Socket(address, port);

            System.out.println("Client is connected!");

            input = new DataInputStream(socket.getInputStream());

            output = new DataOutputStream(socket.getOutputStream());

            sc = new Scanner(System.in);

        } catch (UnknownHostException u) {

            System.out.println(u);

        } catch (IOException i) {

            System.out.println(i);

        }

    }
```

```
String caesarEncrypt(String pt, int k) {
    StringBuilder ct = new StringBuilder();
    for (char c : pt.toCharArray()) {
        if (Character.isUpperCase(c)) {
            int ch = (int) c - 'A';
            ch = (ch + k) % 26;
            char cs = (char) (ch + 65);
            ct.append(cs);
        } else if (Character.isLowerCase(c)) {
            int ch = (int) c - 'a';
            ch = (ch + k) % 26;
            char cs = (char) (ch + 97);
            ct.append(cs);
        } else {
            ct.append(c);
        }
    }
    return ct.toString();
}
public void Work() {
    try {
        System.out.print("Enter the message: ");
        String[] ips = sc.nextLine().trim().split("\\s+");
        System.out.print("Enter the key: ");
        int k = sc.nextInt();
        Vector<String> ct = new Vector<>();
        for (String word : ips) {
            ct.add(caesarEncrypt(word, k));
        }
        String cts = String.join(" ", ct);
        System.out.println("The encrypted text is: " + cts);
```

```
            output.writeInt(k);
            output.writeUTF(cts);
            System.out.println(input.readUTF());
            input.close();
            output.close();
            sc.close();
        } catch (IOException i) {
            System.out.println(i);
        }
    }
    public static void main(String[] args) {
        Caesar_Sender cc = new Caesar_Sender("127.0.0.1", 5431);
        cc.Work();
    }
}
```

Receiver:

```
import java.util.*;
import java.io.*;
import java.net.*;
public class Caesar_Receiver {
    private ServerSocket server = null;
    private Socket socket = null;
    private DataInputStream input = null;
    private DataOutputStream output = null;
    public Caesar_Receiver(int port) {
        try {
            server = new ServerSocket(port);
            socket = server.accept();
            System.out.println("Client is connected!\n");
            input = new DataInputStream(socket.getInputStream());
            output = new DataOutputStream(socket.getOutputStream());
```

```
    } catch (UnknownHostException u) {
        System.out.println(u);
    } catch (IOException i) {
        System.out.println(i);
    }
}
String caesarDecrypt(String ct, int k) {
    StringBuilder pt = new StringBuilder();
    for (char c : ct.toCharArray()) {
        if (Character.isUpperCase(c)) {
            int ch = (int) c - 'A';
            ch = (ch - k) % 26;
            if (ch < 0) {
                ch += 26;
            }
            char cs = (char) (ch + 65);
            pt.append(cs);
        } else if (Character.isLowerCase(c)) {
            int ch = (int) c - 'a';
            ch = (ch - k) % 26;
            if (ch < 0) {
                ch += 26;
            }
            char cs = (char) (ch + 97);
            pt.append(cs);
        } else {
            pt.append(c);
        }
    }
    return pt.toString();
}
```

```
    public void Work() {
        try {
            int k = input.readInt();
            String ip = input.readUTF();
            String[] ips = ip.trim().split("\\s+");
            Vector<String> ct = new Vector<>();
            for (String word : ips) {
                ct.add(caesarDecrypt(word, k));
            }
            System.out.println("The received encrypted text is: " + ip);
            System.out.println("The received key is: " + k);
            String cts = String.join(" ", ct);
            System.out.println("The decrypted text is: " + cts);
            output.writeUTF("Success from receiver!");
            input.close();
            output.close();
        } catch (IOException i) {
            System.out.println(i);
        }
    }
    public static void main(String[] args) {
        Caesar_Receiver cs = new Caesar_Receiver(5431);
        cs.Work();
    }
}
```

Output (Caesar Cipher):

Sender:

```
PS C:\Users\vishw\Coding\Cryptography-Sem-6\Week-1> java .\Caesar_Sender.java
Client is connected!
Enter the message: Meet me.
Enter the key: 3
The encrypted text is: Phhw ph.
Success from receiver!
PS C:\Users\vishw\Coding\Cryptography-Sem-6\Week-1>
```

Receiver:

```
PS C:\Users\vishw\Coding\Cryptography-Sem-6\Week-1> java .\Caesar_Receiver.java
Client is connected!

The received encrypted text is: Phhw ph.
The received key is: 3
The decrypted text is: Meet me.
PS C:\Users\vishw\Coding\Cryptography-Sem-6\Week-1>
```

Code (Vigenère Cipher):

Sender:

```
import java.util.*;

import java.io.*;

import java.net.*;


public class Vignere_Sender {

    private Socket socket = null;

    private DataInputStream input = null;

    private DataOutputStream output = null;

    private Scanner sc = null;

    int pos = 0;


    public Vignere_Sender(String address, int port) {

        try {

            socket = new Socket(address, port);

            System.out.println("Client is connected!");

            input = new DataInputStream(socket.getInputStream());

            output = new DataOutputStream(socket.getOutputStream());

            sc = new Scanner(System.in);

        } catch (UnknownHostException u) {

            System.out.println(u);

        } catch (IOException i) {

            System.out.println(i);

        }

    }


    String vignereEncrpyt(String pt, String k) {

        StringBuilder ct = new StringBuilder();

        int klen = k.length();
```

```
    for (char c : pt.toCharArray()) {
        char keyChar = k.charAt(pos);
        int shift;
        if (Character.isUpperCase(keyChar)) {
            shift = keyChar - 'A';
        } else {
            shift = keyChar - 'a';
        }

        if (Character.isUpperCase(c)) {
            int ch = (int) c - 'A';
            ch = (ch + shift) % 26;
            char cs = (char) (ch + 65);
            ct.append(cs);
            pos = (pos + 1) % klen;
        } else if (Character.isLowerCase(c)) {
            int ch = (int) c - 'a';
            ch = (ch + shift) % 26;
            char cs = (char) (ch + 97);
            ct.append(cs);
            pos = (pos + 1) % klen;
        } else {
            ct.append(c);
        }
    }

    return ct.toString();
}

public void Work() {
    try {
```

```
            System.out.print("Enter the message: ");
            String[] ips = sc.nextLine().trim().split("\\s+");
            System.out.print("Enter the key: ");
            String k = sc.next().trim();
            Vector<String> ct = new Vector<>();
            for (String word : ips) {
                ct.add(vignereEncrpyt(word, k));
            }
            String cts = String.join(" ", ct);
            System.out.println("The encrypted text is: " + cts);
            output.writeUTF(k);
            output.writeUTF(cts);
            System.out.println(input.readUTF());
            input.close();
            output.close();
            sc.close();
        } catch (IOException i) {
            System.out.println(i);
        }
    }

    public static void main(String[] args) {
        Vignere_Sender vs = new Vignere_Sender("127.0.0.1", 5431);
        vs.Work();
    }
}
```

Receiver:

```
import java.util.*;
import java.io.*;
import java.net.*;
```

```
public class Vignere_Receiver {
    private ServerSocket server = null;
    private Socket socket = null;
    private DataInputStream input = null;
    private DataOutputStream output = null;
    int pos = 0;

    public Vignere_Receiver(int port) {
        try {
            server = new ServerSocket(port);
            socket = server.accept();
            System.out.println("Client is connected!\n");
            input = new DataInputStream(socket.getInputStream());
            output = new DataOutputStream(socket.getOutputStream());
        } catch (UnknownHostException u) {
            System.out.println(u);
        } catch (IOException i) {
            System.out.println(i);
        }
    }

    String vignereDecrypt(String ct, String k) {
        StringBuilder pt = new StringBuilder();
        int klen = k.length();

        for (char c : ct.toCharArray()) {
            char keyChar = k.charAt(pos);
            int shift;
            if (Character.isUpperCase(keyChar)) {
                shift = keyChar - 'A';
            } else {
```

```
            shift = keyChar - 'a';
        }

        if (Character.isUpperCase(c)) {
            int ch = (int) c - 'A';
            ch = (ch - shift) % 26;
            if (ch < 0) {
                ch += 26;
            }
            char vr = (char) (ch + 65);
            pt.append(vr);
            pos = (pos + 1) % klen;
        } else if (Character.isLowerCase(c)) {
            int ch = (int) c - 'a';
            ch = (ch - shift) % 26;
            if (ch < 0) {
                ch += 26;
            }
            char vr = (char) (ch + 97);
            pt.append(vr);
            pos = (pos + 1) % klen;
        } else {
            pt.append(c);
        }
    }

    return pt.toString();
}

public void Work() {
    try {
```

```
            String k = input.readUTF();
            String ip = input.readUTF();
            String[] ips = ip.trim().split("\\s+");
            Vector<String> ct = new Vector<>();
            for (String word : ips) {
                ct.add(vignereDecrypt(word, k));
            }
            System.out.println("The received encrypted text is: " + ip);
            System.out.println("The received key is: " + k);
            String cts = String.join(" ", ct);
            System.out.println("The decrypted text is: " + cts);
            output.writeUTF("Success from receiver!");
            input.close();
            output.close();
        } catch (IOException i) {
            System.out.println(i);
        }
    }

    public static void main(String[] args) {
        Vignere_Receiver vr = new Vignere_Receiver(5431);
        vr.Work();
    }
}
```

Output (Vigenère Cipher):

Sender:

```
PS C:\Users\vishw\Coding\Cryptography-Sem-6\Week-1\Vignere Cipher> java .\Vignere_Sender.java
Client is connected!
Enter the message: Geeks for Geeks.
Enter the key: Ayush
The encrypted text is: Gcycz fml Yleim.
Success from receiver!
PS C:\Users\vishw\Coding\Cryptography-Sem-6\Week-1\Vignere Cipher>
```

Receiver:

```
PS C:\Users\vishw\Coding\Cryptography-Sem-6\Week-1\Vignere Cipher> java .\Vignere_Receiver.java
Client is connected!

The received encrypted text is: Gcycz fml Yleim.
The received key is: Ayush
The decrypted text is: Geeks for Geeks.
PS C:\Users\vishw\Coding\Cryptography-Sem-6\Week-1\Vignere Cipher>
```

Code (Vernam Cipher):

Sender:

```
import java.util.*;

import java.io.*;

import java.net.*;


public class Vernam_Sender {

    private Socket socket = null;

    private DataInputStream input = null;

    private DataOutputStream output = null;

    private Scanner sc = null;

    int pos = 0;


    public Vernam_Sender(String address, int port) {

        try {

            socket = new Socket(address, port);

            System.out.println("Client is connected!");

            input = new DataInputStream(socket.getInputStream());

            output = new DataOutputStream(socket.getOutputStream());

            sc = new Scanner(System.in);

        } catch (UnknownHostException u) {

            System.out.println(u);

        } catch (IOException i) {

            System.out.println(i);

        }

    }


    String vernamEncrypt(String pt, String k) {

        StringBuilder ct = new StringBuilder();

        for (int i = 0; i < pt.length(); i++) {

            char pChar = pt.charAt(i);
```

```
        char kChar = k.charAt(i);
        int pc, kc;
        if (Character.isUpperCase(pChar)) {
            pc = (int) pChar - 'A';
        } else {
            pc = (int) pChar - 'a';
        }
        if (Character.isUpperCase(kChar)) {
            kc = (int) kChar - 'A';
        } else {
            kc = (int) kChar - 'a';
        }
        int cc = (pc ^ kc);
        if (Character.isUpperCase(pChar)) {
            char cChar = (char) (cc + 'A');
            ct.append(cChar);
        } else {
            char cChar = (char) (cc + 'a');
            ct.append(cChar);
        }
    }
    return ct.toString();
}

public void Work() {
    try {
        System.out.print("Enter the message: ");
        String ips = sc.next().trim();
        System.out.print("Enter the key: ");
        String k = sc.next().trim();
        String cts = vernamEncrypt(ips, k);
```

```
            System.out.println("The encrypted text is: " + cts);
            output.writeUTF(k);
            output.writeUTF(cts);
            System.out.println(input.readUTF());
            input.close();
            output.close();
            sc.close();
        } catch (IOException i) {
            System.out.println(i);
        }
    }

    public static void main(String[] args) {
        Vernam_Sender vs = new Vernam_Sender("127.0.0.1", 5431);
        vs.Work();
    }
}
```

Receiver:

```
import java.io.*;
import java.net.*;

public class Vernam_Receiver {
    private ServerSocket server = null;
    private Socket socket = null;
    private DataInputStream input = null;
    private DataOutputStream output = null;
    int pos = 0;

    public Vernam_Receiver(int port) {
        try {
            server = new ServerSocket(port);
```

```
        socket = server.accept();
        System.out.println("Client is connected!\n");
        input = new DataInputStream(socket.getInputStream());
        output = new DataOutputStream(socket.getOutputStream());
    } catch (UnknownHostException u) {
        System.out.println(u);
    } catch (IOException i) {
        System.out.println(i);
    }
}

String vernamDecrypt(String ct, String k) {
    StringBuilder pt = new StringBuilder();
    for (int i = 0; i < ct.length(); i++) {
        char cChar = ct.charAt(i);
        char kChar = k.charAt(i);
        int cc, kc;
        if (Character.isUpperCase(cChar)) {
            cc = (int) cChar - 'A';
        } else {
            cc = (int) cChar - 'a';
        }
        if (Character.isUpperCase(kChar)) {
            kc = (int) kChar - 'A';
        } else {
            kc = (int) kChar - 'a';
        }
        int pc = (cc ^ kc);
        if (Character.isUpperCase(cChar)) {
            char pChar = (char) (pc + 'A');
            pt.append(pChar);
```

```
            } else {
                char pChar = (char) (pc + 'a');
                pt.append(pChar);
            }
        }
        return pt.toString();
    }

    public void Work() {
        try {
            String k = input.readUTF();
            String ip = input.readUTF();
            String cts = vernamDecrypt(ip, k);
            System.out.println("The received encrypted text is: " + ip);
            System.out.println("The received key is: " + k);
            System.out.println("The decrypted text is: " + cts);
            output.writeUTF("Success from receiver!");
            input.close();
            output.close();
        } catch (IOException i) {
            System.out.println(i);
        }
    }

    public static void main(String[] args) {
        Vernam_Receiver vr = new Vernam_Receiver(5431);
        vr.Work();
    }
}
```

Output (Vernam Cipher):

Sender:

```
PS C:\Users\vishw\Coding\Cryptography-Sem-6\Week-1\Vernam Cipher> java .\Vernam_Sender.java
Client is connected!
Enter the message: Oak
Enter the key: son
The encrypted text is: ]oh
Success from receiver!
PS C:\Users\vishw\Coding\Cryptography-Sem-6\Week-1\Vernam Cipher>
```

Receiver:

```
PS C:\Users\vishw\Coding\Cryptography-Sem-6\Week-1\Vernam Cipher> java .\Vernam_Receiver.java
Client is connected!

The received encrypted text is: ]oh
The received key is: son
The decrypted text is: Oak
PS C:\Users\vishw\Coding\Cryptography-Sem-6\Week-1\Vernam Cipher>
```